The Quantum Fate of the Good Boussinesq Equation's Resonant and Number-Changing Channels

Project *quantum_boussinesq_solitons* (Ranked next step #3) — PI: Gustav Delius

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Abstract

We investigate ranked next step #3 of the Phase 2 master report (`phase2/report.pdf`): the classical kinematics, hierarchy conservation laws, and semiclassical/quantum implications of the resonant ($A_{12} \to 0$ or ∞), number-changing ($2 \to 1$ merging and $1 \to 2$ splitting Y-junctions), and singular ($A_{12} < 0$ and $A_{\rm br} < 1/4$) soliton channels of the **good** Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$. Every nontrivial claim is labelled **(a)** (proved in the literature), **(b)** (conjectured or heuristic in the literature), **(c)** (derived and verified in this project), or **(d)** (quantum interpretation, conjecture, or open physical question).

Our main structural results **(c)** and quantum conclusions **(d)** are:

1. **KP-triangle unification of resonant kinematics and light-branch stability (`phase2/scripts/resonant_channels_kinematics.py`, `phase2/scripts/resonant_channels_quantum.py`).** In KP root variables, a soliton $(\alpha, \epsilon) \in (0, \pi/2) \times \{\pm 1\}$ carries an ordered chord (p_a, p_b) of the cubic $4p^3 - p = \frac{\ell}{3\sqrt{3}}$ ($\ell = -\epsilon \cos 3\alpha$) on the ellipse $p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}$ **(a)**/**(c)**. Every nondegenerate real KP triple $q_1 > q_2 > q_3$ ($\ell \in (-1, 1)$) spans **two opposite-moving light chords** $S_{12} = (q_1, q_2)$ ($\alpha_{12} \in (\pi/6, \pi/3)$, $\epsilon_{12} = +1$) and $S_{23} = (q_2, q_3)$ ($\alpha_{23} \in (0, \pi/6)$, $\epsilon_{23} = -1$, or with signs reversed) and **one heavy chord** $S_{13} = (q_1, q_3)$ ($\alpha_{13} = \alpha_{12} + \alpha_{23} + \frac{\pi}{3}$ or $\frac{2\pi}{3} - (\alpha_{12} - \alpha_{23}) \in (\pi/3, \pi/2)$) **(c)**. The three genuine codimension-one (measure-zero) resonance curves in $(\alpha_1, \alpha_2) \in (0, \pi/2)^2$—head-on $R_{\rm ho}^+$ ($\alpha_1 + \alpha_2 = \pi/3$, $A_{\rm ho} = \infty$), head-on $R_{\rm ho}^-$ ($|\alpha_1 - \alpha_2| = \pi/3$, $A_{\rm ho} = 0$), and overtaking $R_{\rm ov}$ ($\alpha_1 + \alpha_2 = 2\pi/3$, $A_{\rm ov} = 0$; we confirm `audit_smx.pdf` that $|\alpha_1 - \alpha_2| = \pi/3$ is *not* an overtaking resonance)—are the three pairwise projections of this single family of KP triangles **(c)**. On the light branch $\alpha \in (0, \pi/3)$ ($|P| \in (0, \sqrt{3}/8)$), $E_{\rm light}(P)$ is strictly concave in $|P|$ with $\frac{{\rm d}^2 E_{\rm light}}{{\rm d}P^2} = -\frac{3}{2 \sin 3\alpha} < 0$, so $E_{\rm light}(P_1) + E_{\rm light}(P_2) > E_{\rm light}(P_1 + P_2)$ for all nonzero (P_1, P_2) **(c)**: **no light soliton can split into two solitons ($1_{\rm light} \not\to 2$), and no two light solitons can fuse into a light soliton ($2_{\rm light} \not\to 1_{\rm light}$) (c)**. Conversely, **every heavy soliton ($\alpha \in (\pi/3, \pi/2)$) is embedded in the two-light-soliton head-on continuum** with non-vanishing $1 \to 2$ phase-space Jacobian $J_{1\to 2} = |v_{12} - v_{23}|^{-1} = (\sqrt{3}\, k_{13})^{-1}$ **(c)**.
2. **Confirmation of the KP-chord hypothesis for classical hierarchy charges, and its one-loop breakdown at fixed physical momentum (`phase2/scripts/resonant_channels_conservation.py`).** Using the hyperbolic polynomial map $T = \tanh(kx/2)$, we integrate the Gelfand–Dickey densities ${\rm Res}\, L^{n/3} - {\rm Res}\, L_{\rm vac}^{n/3}$ **symbolically** on the one-soliton profile through spin $n = 8$ ($H_1, H_2, H_3 = 0, H_4, H_5, H_6 = 0, H_7, H_8$, extending Phase 1's numerical fit for $n \le 5$ to exact symbolic proofs through $n = 8$, including the new exact decompositions $H_7 = \Delta_7 - \frac{7}{12}\Delta_5 + \frac{7}{324}\Delta_1$ and $H_8 = \Delta_8 - \frac{7}{36}\Delta_4 + \frac{11}{324}\Delta_2$ with $\Delta_m = (-p_b)^m - (-p_a)^m$) **(c)**. Every physical charge (M, Π, P, E) and every hierarchy charge H_n has the telescopic chord form $Q(p_a, p_b) = F_Q(p_b) - F_Q(p_a)$ for an explicit polynomial $F_Q(p)$ **(c)**. Consequently, chord concatenation $(q_1, q_2) + (q_2, q_3) \to (q_1, q_3)$ satisfies $Q(S_{12}) + Q(S_{23}) = Q(S_{13})$ **identically for the entire classical hierarchy (c)** (verified to 10^{-50} by direct spatial quadrature on the exact 3-exponential Y-junction solution across collision). Unlike in sine-Gordon, the classical conservation laws do **not** forbid $2 \leftrightarrow 1$ fusion **(c)**. At one loop, fixed-profile mode-sum shifts $(\Delta E_1^{\rm ren}, \Delta P_{\rm mode})$ remain telescopic **(c)**, but at fixed physical (dressed) momentum $P^{\rm phys}$ the dispersion shift $F|_P(S) = \Delta E_1^{\rm ren}(S) - v(S) \Delta P_{\rm mode}(S) = -L_1(S)$ breaks the chord

form because $v(S) = \sqrt{3}(p_a + p_b)$ multiplies $\Delta P_{\rm mode}(S)$, yielding a nonzero mismatch $\Delta(F|_P) = F|_P(S_{12}) + F|_P(S_{23}) - F|_P(S_{13}) = 0.38879244070035460274$ at $\phi = 0.23$ **(c)**.

3. **Semiclassical phase, unitarity deficit, and three-body factorisation near resonance (`phase2/scripts/resonant_channels_quantum.py`).** At transverse distance $h \to 0^+$ from any of $R_{\rm ov}, R^+_{\rm ho}, R^-_{\rm ho}$, both $\ell_1 - \ell_2 = c_\ell h + O(h^2)$ and $A_{12} = c_A h^{\sigma_{\rm res}}(1+O(h))$ ($\sigma_{\rm res} = \pm 1$) have simple zeros/poles, so the classical shift $\Delta x_j = \mp k_j^{-1} \ln A_{12}$ diverges as $\mp \sigma_{\rm res} k_j^{-1} \ln h$, while the tree-level phase $\delta(1,2) = T_{12}|_{\rm res} + \frac{2}{9}\sigma_{\rm res} c_\ell\, h \ln h + O(h)$ has a finite limit $T_{12}|_{\rm res}$ with a logarithmic branch-point slope $\partial_h \delta \propto \ln h$ **(c)** (confirming `audit_smx.pdf`). At fixed physical momenta, the $O(\hbar^0)$ Taylor counterterm $-\left(\frac{\Delta P_1}{k_1} - \frac{\Delta P_2}{k_2}\right) \ln A_{12}$ diverges as $\ln h$ on the classical resonance curve **(c)**. Continuing into the singular domain $A_{12} < 0$ gives $\operatorname{Im}\delta = \frac{2\pi}{9}|\ell_1 - \ell_2| > 0$ on the physical sheet, so $|S^{\rm WKB}_{2\to 2}|^2 = \exp(-\frac{4\pi}{9\hbar}|\ell_1 - \ell_2|) < 1$ violates elastic two-body unitarity **(c)**. In 3-soliton collisions with a resonant pair $S_{12} + S_{23} \to S_{13}$, classical cross-ratios and phases factorise exactly ($A_{3,12}A_{3,23} = A_{3,13}$ and $\delta_{\rm cl}(3,S_{12}) + \delta_{\rm cl}(3,S_{23}) = \delta_{\rm cl}(3,S_{13})$) and bare one-loop phonon integrals telescope ($I_{SS}(3,S_{12}) + I_{SS}(3,S_{23}) = I_{SS}(3,S_{13})$), but the continued Gamma candidate fails the bulk fusion bootstrap by $\mathrm{e}^{\mp \mathrm{i}\frac{\pi}{2}\operatorname{sgn}\eta}(1+O(\hbar))$ and the target-soliton momentum renormalisation fails to telescope **(c)**.

4. **Conclusion on elastic factorised scattering ((d) based on (c)).** Purely elastic, unitary, factorised two-soliton scattering is kinematically protected **only in the light-light overtaking sector** ($\alpha_1, \alpha_2 \in (0, \pi/3)$, $\epsilon_1 = \epsilon_2$) and in the sub-threshold **light-light attractive head-on triangle** ($\alpha_1 + \alpha_2 < \pi/3$, $\epsilon_1 = -\epsilon_2$), where $A_{12} > 0$ and no real on-shell $2 \to 1$ or $1 \to 2$ channel exists **(c)**/**(d)**. Heavy solitons ($\alpha \in (\pi/3, \pi/2)$) are quantum mechanically unstable resonance states embedded in the two-light-soliton head-on continuum **(d)**, and at or beyond the resonant boundaries ($A_{12} = 0, \infty$ and $A_{12} < 0$) elastic 2-body unitarity must be augmented by the $2 \leftrightarrow 1$ fusion/splitting channel **(d)**.

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1 Scope, conventions, and audit precedence

We work in the fixed project conventions (`AGENTS.md`, `README.md`, `phase2/report.pdf`) for the **good** Boussinesq equation **(a)**:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \varphi_x = 2(\ln\tau)_{xx}, \qquad (D_t^2 - D_x^2 + D_x^4)\,\tau\cdot\tau = 0, \tag{1}$$

with Lagrangian density $\mathcal{L} = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2$ and canonical symplectic form $\Omega = \int \mathrm{d}\pi\wedge\mathrm{d}\varphi$ ($\pi = \varphi_t$). A one-soliton with amplitude parameter $\alpha \in (0,\pi/2)$ and direction $\epsilon \in \{+1,-1\}$ has **(a)**/**(c)**:

$$\begin{aligned} &k = \sin\alpha, \quad v = \epsilon\cos\alpha, \quad \omega = kv = \frac{\epsilon}{2}\sin 2\alpha, \quad \ell = -\epsilon\cos 3\alpha, \\ &M = 2k, \quad \Pi = -Mv = -2\epsilon k\cos\alpha, \quad P = \frac{2}{3}\epsilon k^3\cos\alpha, \quad E = \frac{2}{15}k^3(5-4k^2), \end{aligned} \tag{2}$$

satisfying $\frac{\mathrm{d}E}{\mathrm{d}P} = v$ and $\frac{\mathrm{d}P}{\mathrm{d}\alpha} = \frac{2}{3}\epsilon k\sin 3\alpha = \frac{2}{9}k\,\ell'(\alpha)$ **(c)**. At the momentum cusp $\alpha = \pi/3$ ($k = \sqrt{3}/2$, $|v| = 1/2$, $|\ell| = 1$), $|P|$ attains its global maximum $P_{\max} = \sqrt{3}/8 \approx 0.21650635$ and $\frac{\mathrm{d}P}{\mathrm{d}\alpha} = 0$, dividing the one-soliton manifold into the **light branch** $\alpha \in (0,\pi/3)$ ($k < \sqrt{3}/2$, $|v| > 1/2$) and the **heavy branch** $\alpha \in (\pi/3,\pi/2)$ ($k > \sqrt{3}/2$, $|v| < 1/2$) **(c)** (`phase1/review.pdf`).

Audit precedence. Following the ground rules of the task, audit verdicts override package claims. In particular, we incorporate the corrections from `audit_smx.pdf`, `audit_defq.pdf`, `audit_det_projected.pdf`, and `errata.pdf`:

- The line $|\alpha_1 - \alpha_2| = \pi/3$ is *not* an overtaking resonance (`audit_smx.pdf`); we verify this directly in `phase2/scripts/resonant_channels_kinematics.py`.
- The tree-level S-matrix limit at a genuine resonance is continuous, not differentiably (C^1) smooth (`audit_smx.pdf`); we derive the exact $h\ln h$ coefficient on all three genuine resonance lines in `phase2/scripts/resonant_channels_quantum.py`.
- Every two-phase zero-background Hirota breather of the good equation has $A_{\mathrm{br}} = \frac{1-3\cot^2 a}{1+3\coth^2 b} < \frac{1}{4} < 1$ and is singular (`phase1/review.pdf`, `audit_smx.pdf`); this does not by itself exclude quantum bound states beyond the two-phase Hirota ansatz **(d)**.
- In `audit_defq.pdf`, the classical KP-edge charge factorisation survives, while the canonical-charge triangle holonomy no-go theorem is qualified as conditional on fixed unshifted absorption rapidities and single-valued vertex counterterms.

2 Question 1: Classical kinematics of resonant and number-changing channels

2.1 KP root/chord dictionary and the three genuine resonance loci

In the third-order Lax representation $L = \partial_x^3 + (\frac{3}{2}u - \frac{1}{4})\partial_x + r$ **(a)** [10, 4], a soliton (α,ϵ) corresponds to an ordered pair (chord) (p_a, p_b) ($p_a > p_b$) of real roots of the KP spectral cubic **(a)**/**(c)** (`phase1/review.pdf`, `errata.pdf` M06–M07, `phase2/scripts/resonant_channels_kinematics.py`):

$$p_a = \frac{1}{2}\left(\frac{v}{\sqrt{3}} + k\right) = \frac{\epsilon}{\sqrt{3}}\cos\left(\alpha - \frac{\epsilon\pi}{3}\right), \qquad p_b = \frac{1}{2}\left(\frac{v}{\sqrt{3}} - k\right) = \frac{\epsilon}{\sqrt{3}}\cos\left(\alpha + \frac{\epsilon\pi}{3}\right), \tag{3}$$

so that $p_a - p_b = k > 0$, $p_a + p_b = \frac{v}{\sqrt{3}}$, and both roots satisfy

$$4p^3 - p = \frac{\ell}{3\sqrt{3}} = -\frac{\epsilon}{3\sqrt{3}}\cos 3\alpha, \qquad p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}. \tag{4}$$

The Hirota two-soliton interaction coefficient $A_{12} = -\frac{\mathcal{P}(k_1-k_2,\omega_1-\omega_2)}{\mathcal{P}(k_1+k_2,\omega_1+\omega_2)}$ ($\mathcal{P}(K,W) = W^2 - K^2 + K^4$) equals the KP cross-ratio in both relative orientations **(a)** [5], **(c)** (`phase2/scripts/resonant_channels_kinematics.py`):

$$A_{12} = \frac{(p_{1a}-p_{2a})(p_{1b}-p_{2b})}{(p_{1a}-p_{2b})(p_{1b}-p_{2a})} = \begin{cases} \dfrac{1-3\cot^2\left(\frac{\alpha_1+\alpha_2}{2}\right)}{1-3\cot^2\left(\frac{\alpha_1-\alpha_2}{2}\right)} & (\text{overtaking},\ \epsilon_1=\epsilon_2), \\ \dfrac{1-3\tan^2\left(\frac{\alpha_1-\alpha_2}{2}\right)}{1-3\tan^2\left(\frac{\alpha_1+\alpha_2}{2}\right)} & (\text{head-on},\ \epsilon_1=-\epsilon_2). \end{cases} \tag{5}$$

Proposition 2.1 (Exact resonance loci and KP-triangle unification, **(c)**, `phase2/scripts/resonant_channels_kinematics.py`). *Let $(\alpha_1,\alpha_2) \in Q := (0,\pi/2)^2$ with $\alpha_1 \neq \alpha_2$ when $\epsilon_1 = \epsilon_2$.*

1. ***Characterization by shared KP roots and equal ℓ:*** *A resonant interaction $A_{12} = 0$ or $A_{12} = \infty$ occurs if and only if solitons 1 and 2 share a root of the KP cubic (4), which in $Q = (0,\pi/2)^2$ implies $\ell_1 = \ell_2$ and defines three real-analytic codimension-one lines:*

   - Overtaking resonance R_{ov} ($\epsilon_1 = \epsilon_2$, $A_{\text{ov}} = 0$):

   $$\alpha_1 + \alpha_2 = \frac{2\pi}{3}, \qquad \alpha_1, \alpha_2 \in \left(\frac{\pi}{6}, \frac{\pi}{2}\right), \tag{6}$$

   *where $p_{1a} = p_{2a}$ (for $\epsilon_1 = \epsilon_2 = +1$) or $p_{1b} = p_{2b}$ (for $\epsilon_1 = \epsilon_2 = -1$), and $\ell_1 = \ell_2 = -\epsilon_1 \cos 3\alpha_1 > 0$. The overtaking denominator $1 - 3\cot^2\left(\frac{\alpha_1-\alpha_2}{2}\right)$ vanishes only at $|\alpha_1 - \alpha_2| = \frac{2\pi}{3} \notin Q$, so A_{ov} has **no pole** in Q. In particular, on the false line $|\alpha_1 - \alpha_2| = \frac{\pi}{3}$ (refuted in `audit_smx.pdf`), the denominator equals $1 - 3\cot^2(\pi/6) = -8 \neq 0$; at $(\alpha_1,\alpha_2) = (0.2, 0.2 + \pi/3)$ we have $A_{\text{ov}} = 0.35546547984906164962$ and $\ell_1 - \ell_2 = -2\cos(0.6) = -1.6506712298193565945 \neq 0$.*

   - Head-on attractive resonance R^{+}_{ho} ($\epsilon_1 = -\epsilon_2$, $A_{\text{ho}} = \infty$):

   $$\alpha_1 + \alpha_2 = \frac{\pi}{3}, \qquad \alpha_1, \alpha_2 \in \left(0, \frac{\pi}{3}\right), \tag{7}$$

   where $p_{1b} = p_{2a}$ (for $\epsilon_1 = +1, \epsilon_2 = -1$) or $p_{1a} = p_{2b}$ (for $\epsilon_1 = -1, \epsilon_2 = +1$), and $\ell_1 = \ell_2 = -\epsilon_1 \cos 3\alpha_1$.

   - Head-on repulsive resonance R^{-}_{ho} ($\epsilon_1 = -\epsilon_2$, $A_{\text{ho}} = 0$):

   $$|\alpha_1 - \alpha_2| = \frac{\pi}{3}, \qquad \min(\alpha_1,\alpha_2) \in \left(0, \frac{\pi}{6}\right), \quad \max(\alpha_1,\alpha_2) \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right), \tag{8}$$

   where $p_{1b} = p_{2b}$ (for $\alpha_1 - \alpha_2 = \pi/3$, $\epsilon_1 = +1, \epsilon_2 = -1$) or $p_{1a} = p_{2a}$ (for $\alpha_2 - \alpha_1 = \pi/3$, $\epsilon_1 = +1, \epsilon_2 = -1$), and $\ell_1 = \ell_2$ because $\cos(3\alpha_2 \pm \pi) = -\cos 3\alpha_2$ compensates $\epsilon_1 = -\epsilon_2$.

2. ***Unification via KP triangles and light/heavy branch assignment:*** *Every nondegenerate KP cubic $4q^3 - q = \frac{\ell}{3\sqrt{3}}$ with $\ell \in (-1,1)$ has three real roots ordered $q_1 > q_2 > q_3$, parametrised by $\phi \in (0,\pi/3)$ as*

   $$\begin{aligned} q_1 &= \frac{1}{\sqrt{3}}\cos\phi \in \left(\frac{1}{2\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \qquad q_2 = \frac{1}{\sqrt{3}}\cos\left(\phi - \frac{2\pi}{3}\right) \in \left(-\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}\right), \\ q_3 &= \frac{1}{\sqrt{3}}\cos\left(\phi + \frac{2\pi}{3}\right) \in \left(-\frac{1}{\sqrt{3}}, -\frac{1}{2\sqrt{3}}\right). \end{aligned} \tag{9}$$

The three chords $S_{12} = (q_1, q_2)$, $S_{23} = (q_2, q_3)$, and $S_{13} = (q_1, q_3)$ have wavenumbers and velocities:

$$\begin{aligned} S_{12}:\ & k_{12} = q_1 - q_2 = \sin\Big(\frac{\pi}{3} - \phi\Big) < \frac{\sqrt{3}}{2}, \quad v_{12} = -\sqrt{3}\, q_3 = +\cos\Big(\frac{\pi}{3} - \phi\Big) > \frac{1}{2} \quad (\textit{right } \textbf{\textit{light}}), \\ S_{23}:\ & k_{23} = q_2 - q_3 = \sin\phi < \frac{\sqrt{3}}{2}, \qquad\qquad v_{23} = -\sqrt{3}\, q_1 = -\cos\phi < -\frac{1}{2} \quad (\textit{left } \textbf{\textit{light}}), \\ S_{13}:\ & k_{13} = q_1 - q_3 = \sin\Big(\phi + \frac{\pi}{3}\Big) > \frac{\sqrt{3}}{2}, \quad v_{13} = -\sqrt{3}\, q_2 = \cos\Big(\phi + \frac{\pi}{3}\Big) \in \Big(-\frac{1}{2}, \frac{1}{2}\Big) \quad (\textbf{\textit{heavy}},\ \phi \neq \tfrac{\pi}{6}). \end{aligned} \tag{10}$$

For $\phi \in (0, \pi/6)$ ($\ell > 0$), S_{13} is a right-moving heavy soliton with $\alpha_{13} = \phi + \frac{\pi}{3} \in (\pi/3, \pi/2)$, and the three pairs of chords of the same KP triangle $\{S_{12}, S_{23}, S_{13}\}$ project precisely onto the three resonance lines:

$$(S_{12}, S_{23}) \in R_{\mathrm{ho}}^{+} \ (\textit{light–light}), \quad (S_{13}, S_{23}) \in R_{\mathrm{ho}}^{-} \ (\textit{heavy–light}), \quad (S_{12}, S_{13}) \in R_{\mathrm{ov}} \ (\textit{light–heavy}). \tag{11}$$

(For $\phi \in (\pi/6, \pi/3)$, $\ell < 0$ and S_{13} is a left-moving heavy soliton with $\alpha_{13} = \frac{2\pi}{3} - \phi \in (\pi/3, \pi/2)$, giving the mirror-image projections.)

2.2 Reachability from real asymptotic states and domain measure

On any of the three resonance curves, dropping the vanishing or divergent four-exponential term (after shifting θ_j when $A_{12} \to \infty$) yields the exact three-exponential Hirota Y-junction **(a)** [7, 8, 9, 5, 3]:

$$\tau_Y(x,t) = c_1\, \mathrm{e}^{\eta_1} + c_2\, \mathrm{e}^{\eta_2} + c_3\, \mathrm{e}^{\eta_3}, \qquad \eta_i = \pm\big(q_i x - \sqrt{3}\, q_i^2 t\big) + \eta_{i,0} \quad (c_i > 0). \tag{12}$$

Because $\mathcal{P}\big(q_i - q_j, \sqrt{3}(q_i^2 - q_j^2)\big) = 4(q_i - q_j)^2\big(q_i^2 + q_i q_j + q_j^2 - \frac{1}{4}\big) = 0$ for every pair (i, j) of a KP triple (9), $\tau_Y(x,t) > 0$ solves $(D_t^2 - D_x^2 + D_x^4)\tau_Y \cdot \tau_Y = 0$ identically and $u_Y = 2(\ln \tau_Y)_{xx}$ is globally smooth for all $(x,t) \in \mathbb{R}^2$ **(a)**/**(c)** (verified to 10^{-50} against the nonlinear PDE in `phase2/scripts/resonant_channels_kinematics.py`).

Proposition 2.2 (Asymptotic reachability and measure of resonant/singular domains, **(c)**, `phase2/scripts/resonant_channels_kinematics.py`)**.** *Let $q_1 > q_2 > q_3$ be a nondegenerate KP triple with velocities $v_{12} > v_{13} > v_{23}$ as in (10).*

1. ***Exact** $1 \to 2$ **splitting and** $2 \to 1$ **merging asymptotics:** Along the ray $x = v_{13} t + z$, $\eta_1 - \eta_3 = \pm(q_1 - q_3) z$ is stationary while*

$$\eta_2 - \eta_1 = \pm\sqrt{3}\,(q_1 - q_2)(q_2 - q_3)\, t + O(z) = \pm\sqrt{3}\, k_{12} k_{23}\, t + O(z). \tag{13}$$

Since $k_{12} k_{23} > 0$:

- *For the $+$ sign in (12), as $t \to -\infty$ the middle exponential e^{η_2} is suppressed by $\mathrm{e}^{-\sqrt{3} k_{12} k_{23} |t|}$ and $u_Y(x,t)$ approaches a **single heavy soliton** S_{13} moving along $x = v_{13} t$; as $t \to +\infty$, $u_Y(x,t)$ splits ($1 \to 2$) into the **two well-separated light solitons** S_{12} (moving right at $v_{12} > 1/2$) and S_{23} (moving left at $v_{23} < -1/2$).*
- *For the $-$ sign in (12) (time reversal $t \mapsto -t, x \mapsto -x$), as $t \to -\infty$ the solution consists of **two real, smooth, exponentially separated incoming light solitons** S_{12} (at $x \sim v_{12} t < 0$) and S_{23} (at $x \sim v_{23} t > 0$) colliding head-on and **merging** ($2 \to 1$) into the single heavy soliton S_{13} as $t \to +\infty$ (verified at $t = \pm 400$ to 10^{-48} in `phase2/scripts/resonant_channels_kinematics.py`).*

Thus the resonant $2 \to 1$ *and* $1 \to 2$ *channels are* ***directly reachable from real, non-singular asymptotic soliton states****. What appears as an overtaking resonance* $R_{\rm ov}$ *(*$S_{12} + S_{13} \leftrightarrow S_{23}$*) or repulsive head-on resonance* $R^-_{\rm ho}$ *(*$S_{13} + S_{23} \leftrightarrow S_{12}$*) in two-soliton coordinates is the* same *spatial Y-junction viewed by pairing the heavy leg* S_{13} *with one of the two light legs.*

2. ***Measure zero of resonant loci vs. positive measure of singular regions:***

   - *The resonant loci* $R_{\rm ov}$*,* $R^+_{\rm ho}$*, and* $R^-_{\rm ho}$ *are smooth 1D curves in the 2D kinematic square* $Q = (0, \pi/2)^2$ *(and in the four (light/heavy)* $\times$ *(light/heavy) sheets over* $(|P_1|, |P_2|) \in (0, \sqrt{3}/8)^2$*), hence of* ***two-body Lebesgue measure zero*** *(codimension* 1*).*
   - *However, in the* ***one-body*** *kinematic domain of the heavy branch* $\alpha_{13} \in (\pi/3, \pi/2)$*, the* $1 \to 2$ *decay channel* $S_{13} \to S_{12} + S_{23}$ *is open on the* ***entire heavy branch*** *(full 1D measure, codimension* 0*)!*
   - *In the 2D two-body square* $Q = (0, \pi/2)^2$ *(area* $\pi^2/4$*), the open singular regions (*$A_{12} < 0$*, where the 4-exponential Hirota solution has logarithmic poles* $\tau = 0$*) have* ***positive area****:* $\frac{2}{9}$ *of* Q *for overtaking (*$\alpha_1 + \alpha_2 > 2\pi/3$*) and* $\frac{2}{3}$ *of* Q *for head-on (*$\alpha_1 + \alpha_2 > \pi/3$ *with* $|\alpha_1 - \alpha_2| < \pi/3$*). Crucially, inside the* ***light-light square*** $(0, \pi/3)^2$ *(area* $\pi^2/9$*,* $\frac{4}{9}$ *of* Q*),* 100% ***of overtaking collisions*** *(*$\alpha_1 + \alpha_2 < 2\pi/3$*) and* 50% ***of head-on collisions*** *(*$\alpha_1 + \alpha_2 < \pi/3$*) are strictly regular (*$A_{12} > 0$*) and non-resonant* ***(c)****.*

3 Question 2: Conservation laws and the KP-chord hypothesis

3.1 Exact symbolic proof of the chord form through spin 8

In the Gelfand–Dickey hierarchy **(a)** [4], the conserved charges are $H_n = \int_{-\infty}^{\infty} \big(\operatorname{Res} L^{n/3} - \operatorname{Res} L_{\rm vac}^{n/3} \big)\, \mathrm{d}x$, where $L = \partial_x^3 + q\,\partial_x + r$ has $q = \frac{3}{2}u - \frac{1}{4}$ and $r = \frac{1}{2}q_x - \frac{v}{2\sqrt{3}}(q + \frac{1}{4})$ on a soliton moving at velocity v. For $n = 3m$, L^m is a purely differential operator so $\operatorname{Res} L^m \equiv 0$ identically **(a)**. In Phase 1 (`phase1/scripts/charges.py`), H_n was integrated numerically for $n \in \{1, 2, 4, 5\}$ and fitted to linear combinations of $\Delta_m := \mu_1^m - \mu_2^m = (-p_b)^m - (-p_a)^m$. Here we test the KP-chord hypothesis **symbolically** through $n = 8$ by mapping the one-soliton profile to a polynomial in $T = \tanh(kx/2) \in (-1, 1)$ **(c)** (`phase2/scripts/resonant_channels_conservation.py`).

Theorem 3.1 (Symbolic chord-potential structure of the Boussinesq hierarchy, **(c)**, `phase2/scripts/resonant_channels_conservation.py`)**.** *Let* (p_a, p_b) *(*$p_a > p_b$*) be the KP chord of a soliton* (α, ϵ) *on* $p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}$*, and define* $\Delta_m := (-p_b)^m - (-p_a)^m = (-1)^{m-1}(p_a^m - p_b^m)$*.*

1. ***Exact symbolic integration of*** $H_1, \ldots, H_8$***:*** *Under* $T = \tanh(kx/2)$ *(*$u = \frac{k^2}{2}(1 - T^2)$*,* $\partial_x = \frac{k}{2}(1 - T^2)\frac{\mathrm{d}}{\mathrm{d}T}$*,* $\mathrm{d}x = \frac{2}{k(1-T^2)}\,\mathrm{d}T$*), the vacuum-subtracted densities* $\operatorname{Res} L^{n/3} - \operatorname{Res} L_{\rm vac}^{n/3}$ *reduce to exact polynomials in* T *whose integrals over* $T \in [-1, 1]$ *decompose modulo* $p_a^2 + p_a p_b + p_b^2 - \frac{1}{4} = 0$ *as:*

$$
\begin{aligned}
&H_1 = \Delta_1, && H_2 = \Delta_2, && H_3 = 0,\\
&H_4 = \Delta_4 - \frac{1}{3}\Delta_2, && H_5 = \Delta_5 - \frac{5}{72}\Delta_1, && H_6 = 0, \qquad (14)\\
&H_7 = \Delta_7 - \frac{7}{12}\Delta_5 + \frac{7}{324}\Delta_1, && H_8 = \Delta_8 - \frac{7}{36}\Delta_4 + \frac{11}{324}\Delta_2.
\end{aligned}
$$

2. ***Univariate chord potentials*** $Q(p_a, p_b) = F_Q(p_b) - F_Q(p_a)$***:*** *Every hierarchy charge* H_n *(*$1 \le n \le 8$*) and every physical charge* M, Π, P, E *has the exact chord-difference form*

$Q(p_a,p_b) = F_Q(p_b) - F_Q(p_a)$ *modulo* $p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}$, *with univariate polynomials:*

$$\begin{aligned} F_M(p) &= -2p, & F_\Pi(p) &= 2\sqrt{3}\,p^2, \\ F_P(p) &= -2\sqrt{3}\,p^4 + \frac{1}{\sqrt{3}}\,p^2, & F_E(p) &= -\frac{24}{5}\,p^5 + \frac{1}{6}\,p, \\ F_{H_7}(p) &= -p^7 + \frac{7}{12}\,p^5 - \frac{7}{324}\,p, & F_{H_8}(p) &= p^8 - \frac{7}{36}\,p^4 + \frac{11}{324}\,p^2. \end{aligned} \tag{15}$$

3. ***Why all hierarchy charges have the chord form, and the spin-$3m$ gap:*** *On the zero-background KP hierarchy ($\tilde{L} = \partial_x^3 + \tilde{q}\partial_x + \tilde{r}$, $\tilde{q}_\infty = 0$), a line soliton with chord (μ_1,μ_2) is generated by a Sato dressing transformation shifting* $\ln\tau$ *by* $\ln\frac{\partial-\mu_1}{\partial-\mu_2}$, *so* $\int \operatorname{Res}\tilde{L}^{n/3}\,\mathrm{d}x = \mu_1^n - \mu_2^n = \Delta_n$ ***(a)*** *[4]. Shifting to the good Boussinesq constant background $L = \tilde{L} - \frac{1}{4}\partial_x$ mixes $\tilde{L}^{n/3}$ only with lower powers $\tilde{L}^{(n-2j)/3}$ of the same parity, so every hierarchy charge H_n on a one-soliton is a linear combination of Δ_m ($m \le n$, $m \equiv n \pmod 2$) and therefore has the chord form $F_{H_n}(p_b) - F_{H_n}(p_a)$* ***(a)/(c)****. Crucially, however,* not every *polynomial in (k,v) is a chord difference of matching degree: because $\Delta_3 = \frac{1}{4}\Delta_1$ on $p_a^2 + p_a p_b + p_b^2 = \frac{1}{4}$, the space* $\operatorname{span}\{\Delta_1,\Delta_5\}$ *contains $k^3(5-4k^2) = 36\big(\Delta_5 - \frac{5}{144}\Delta_1\big)$ (the exact functional form of E!), but does* ***not*** *contain pure k^3* ***(c)*** *(`phase2/scripts/resonant_channels_conservation.py`).*

3.2 Telescoping in $2 \leftrightarrow 1$ fusion and one-loop fate at fixed momentum

Proposition 3.2 (Classical conservation vs. one-loop fixed-momentum non-telescoping, **(c)**, `phase2/scripts/resonant_channels_conservation.py`). *Let $q_1 > q_2 > q_3$ be any KP triple with chords $S_{12} = (q_1,q_2)$, $S_{23} = (q_2,q_3)$, and $S_{13} = (q_1,q_3)$.*

1. ***Exact classical conservation of the infinite hierarchy:*** *For every charge Q of the chord form $Q(p_a,p_b) = F_Q(p_b) - F_Q(p_a)$ (including M,Π,P,E and all H_n), chord concatenation $(q_1,q_2) + (q_2,q_3) \to (q_1,q_3)$ telescopes identically:*

$$Q(S_{12}) + Q(S_{23}) = \big(F_Q(q_2) - F_Q(q_1)\big) + \big(F_Q(q_3) - F_Q(q_2)\big) = F_Q(q_3) - F_Q(q_1) = Q(S_{13}). \tag{16}$$

   Direct 60-digit spatial quadrature of the nonlinear charge densities of M,Π,P,E on the exact 3-exponential Y-junction solution $u_Y(x,t)$ at $t \in \{-2,0,+2\}$ agrees with $Q(S_{12}) + Q(S_{23}) = Q(S_{13})$ to $< 10^{-50}$ ***(c)****. Thus* ***the classical conservation laws do not forbid $2 \leftrightarrow 1$ soliton fusion or splitting****, in sharp contrast to sine-Gordon (where kinks carry a fixed topological charge and a 1-parameter dispersion $Q_s = m^s \sinh(s\theta)$, so $Q_s(\theta_1) + Q_s(\theta_2) = Q_s(\theta_3)$ has no real solution).*

2. ***Equivalence to the defect KP-triangle holonomy and one-loop breakdown at fixed P^{phys}:*** *For a Corrigan–Zambon defect of orientation $\eta \in \{\pm 1\}$ on the KP triangle $\{q_1,q_2,q_3\}$ ($q_1 > q_2 > q_3$), the absorption signs on the three directed edges $(q_1 \to q_2, q_2 \to q_3, q_3 \to q_1)$ are $\rho = (+\eta,+\eta,-\eta)$ (`defq_charges_fusion.pdf`, `audit_defq.pdf`). Hence the directed KP-triangle holonomy of any soliton observable $\mathcal{O}$ is literally the $2 \to 1$ fusion balance:*

$$\sum_{i=1}^{3} \rho_i\,\mathcal{O}(S_i) = \eta\,\Big(\mathcal{O}(S_{12}) + \mathcal{O}(S_{23}) - \mathcal{O}(S_{13})\Big). \tag{17}$$

   Consequently:

   - *In the* fixed-profile mode-sum *one-loop prescription, $\Delta E_1^{\mathrm{ren}}(S)$ and $\Delta P_{\mathrm{mode}}(S)$ are 2-root dipole differences on the KP cubic and telescope to zero ($\Delta E_1^{\mathrm{ren}}(S_{12}) + \Delta E_1^{\mathrm{ren}}(S_{23}) - \Delta E_1^{\mathrm{ren}}(S_{13}) = 0$ to 10^{-61} at $\phi = 0.23$)* ***(c)****.*

- *However, in the* fixed-physical-momentum *expansion* $E^{\rm phys}(P^{\rm phys}) = E_{\rm cl}(P^{\rm phys}) + \hbar F|_P(P^{\rm phys}) + O(\hbar^2)$*, the charge-pair-invariant one-loop shift* $F|_P(S) = \Delta E_1^{\rm ren}(S) - v(S)\,\Delta P_{\rm mode}(S) = -L_1(S)$ *(`audit_det_projected.pdf`) is* ***not*** *a chord difference because* $v(S) = \sqrt{3}(p_a + p_b)$ *multiplies* $\Delta P_{\rm mode}(S)$*:*

$$\begin{aligned}\Delta(F|_P) &:= F|_P(S_{12}) + F|_P(S_{23}) - F|_P(S_{13}) \\ &= -(v_{12} - v_{13})\,\Delta P_{\rm mode}(S_{12}) - (v_{23} - v_{13})\,\Delta P_{\rm mode}(S_{23}) \neq 0. \end{aligned} \tag{18}$$

At $\phi = 0.23$*,* $\Delta(F|_P) = +0.38879244070035460274 \neq 0$ ***(c)*** *(`phase2/scripts/resonant_channels_conservation.py`).*

Remark 3.3 (Interpretation for quantum fusion, **(d)**). For a single charge pair $(E^{\rm phys}, P^{\rm phys})$, a nonzero one-loop shift $\Delta(F|_P) \neq 0$ does *not* by itself forbid $2 \to 1$ fusion: because $v_{12} \neq v_{13}$ and $v_{23} \neq v_{13}$, shifting the resonant momenta by $P_j^{\rm phys} = P_j^{\rm cl} + \hbar\,\delta P_j$ with $(v_{12} - v_{13})\delta P_{12} + (v_{23} - v_{13})\delta P_{23} = -\Delta(F|_P)$ (for instance $\delta P_j = \Delta P_{\rm mode}(S_j)$, corresponding to unshifted bare KP roots) restores $E^{\rm phys}(P_1^{\rm phys}) + E^{\rm phys}(P_2^{\rm phys}) = E^{\rm phys}(P_1^{\rm phys} + P_2^{\rm phys})$ to $O(\hbar)$ along a shifted 1D curve **(c)**/**(d)**. However, for the *entire infinite tower* of quantum charges $\{\hat{H}_n\}$ to allow $2 \to 1$ fusion on the *same* shifted curve in $(P_1^{\rm phys}, P_2^{\rm phys})$, all one-loop charge corrections $\Delta H_n|_P$ would have to be absorbed by a single renormalisation of the KP triple $\{q_1, q_2, q_3\}$. Whether the quantum Boussinesq hierarchy preserves this exact KP-chord elasticity at one loop and beyond remains open **(d)**.

4 Question 3: Quantum and semiclassical implications

4.1 Semiclassical phase $\delta(1,2)$ and divergent time delay as $A_{12} \to 0$ or ∞

Recall the audited Darboux-frame semiclassical two-soliton phase **(c)** (`phase2/report.pdf` Prop. 2.3, `audit_smx.pdf`):

$$\delta(1,2) = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12} + T_{12}, \qquad T_{12} = \frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1), \tag{19}$$

whose canonical momentum derivatives give the Darboux shifts $\Delta X_1' = -\frac{\partial\delta}{\partial P_1} = -k_1^{-1}\ln A_{12} + \gamma_1 M_2$ ($\gamma_j = \frac{{\rm d}\Pi_j/{\rm d}\alpha_j}{{\rm d}P_j/{\rm d}\alpha_j}$).

Proposition 4.1 (Transverse $h\ln h$ branch point and divergent delay at resonance, **(c)**, `phase2/scripts/resonant_channels_quantum.py`)**.** *Let* $h \to 0^+$ *be the transverse rapidity displacement into the regular domain* ($A_{12} > 0$) *from any of the three genuine resonance lines* $R_{\rm ov}$ ($\alpha_2 = \frac{2\pi}{3} - \alpha_1 - h$, $\sigma_{\rm res} = +1$)*,* $R_{\rm ho}^+$ ($\alpha_2 = \frac{\pi}{3} - \alpha_1 - h$, $\sigma_{\rm res} = -1$)*, or* $R_{\rm ho}^-$ ($\alpha_1 = \alpha_2 + \frac{\pi}{3} + h$, $\sigma_{\rm res} = +1$)*.*

1. ***Simple zero/pole of*** A_{12} ***and linear vanishing of*** $\ell_1 - \ell_2$***:*** *On all three lines, one KP root difference* $p_{1r} - p_{2s} = O(h)$ *vanishes to first order, so both* $\ell_1 - \ell_2$ *and* $A_{12}^{\sigma_{\rm res}}$ *vanish linearly in* h*:*

$$\ell_1 - \ell_2 = c_\ell(\alpha_*)\,h + O(h^2), \qquad A_{12}(h) = c_A(\alpha_*)\,h^{\sigma_{\rm res}}\big(1 + O(h)\big) \qquad (c_A > 0), \tag{20}$$

with $c_\ell(\alpha_*) = -3\sin 3\alpha_* \neq 0$ *for* $\alpha_* \neq \pi/3$*. Consequently, as* $h \to 0^+$*:*

$$\boxed{\delta(1,2) = T_{12}\big|_{\rm res} + \frac{2}{9}\sigma_{\rm res}\,c_\ell(\alpha_*)\,h\ln h + O(h), \qquad \Delta x_1 = -\frac{\sigma_{\rm res}}{k_1}\ln h - \frac{\ln c_A}{k_1} + O(h).} \tag{21}$$

Thus $\delta(1,2)$ *approaches the finite topological limit* $T_{12}|_{\rm res}$ *continuously, while its transverse slope* $\partial_h\delta \sim \frac{2}{9}\sigma_{\rm res}c_\ell\ln h$ *and the classical position shift* Δx_j *(and Wigner time delay* $\Delta t_{12} = \frac{\Delta x_1 - \Delta x_2}{v_1 - v_2}$*)* ***diverge logarithmically as*** $\ln(1/h)$ ***(c)****.*

2. ***Logarithmic divergence of the fixed-physical-momentum $O(\hbar^0)$ phase:*** *While the bare fixed-rapidity one-loop phonon integral $I_{12}(\alpha_1,\alpha_2)$ remains finite on all three resonance lines (`oneloop_phase.pdf`, `audit_smx.pdf`), expressing the S-matrix at fixed physical momenta $P_j^{\rm phys} = P_j^{\rm cl} + \hbar\,\Delta P_j$ introduces the $O(\hbar^0)$ chain-rule counterterm*

$$\delta_{12,\rm renorm} = -\frac{\partial\delta_{\rm cl}}{\partial P_1}\Delta P_1 - \frac{\partial\delta_{\rm cl}}{\partial P_2}\Delta P_2 = -\left(\frac{\Delta P_1}{k_1} - \frac{\Delta P_2}{k_2}\right)\ln A_{12} + \gamma_1 M_2 \Delta P_1 - \gamma_2 M_1 \Delta P_2, \quad (22)$$

*whose coefficient $-\left(\frac{\Delta P_1}{k_1} - \frac{\Delta P_2}{k_2}\right)$ is generically non-zero on all three resonance lines, producing a **logarithmic divergence** $\propto \ln h$ **at** $O(\hbar^0)$ **(c)**:*

Resonance point	$T_{12}\vert_{\rm res}$	c_A	$\frac{2}{9}\sigma_{\rm res}c_\ell$ $(h\ln h)$	$-\left(\frac{\Delta P_1}{k_1} - \frac{\Delta P_2}{k_2}\right)$ $(\ln A_{12})$
$R_{\rm ov}$ $(\alpha_1 = 0.75)$	+0.898441539345	0.074538040233	−0.518715464592	+0.392932242573
$R_{\rm ho}^{+}$ $(\alpha_1 = 0.40)$	+1.076226034357	0.412963868475	+0.621359390645	+1.271224247491
$R_{\rm ho}^{-}$ $(\alpha_2 = 0.25)$	+0.787088643179	1.240490193422	−0.454425840016	+0.992253053483

Remark 4.2 (Physical meaning of the $\ln h$ divergence at $O(\hbar^0)$, **(c)**/**(d)**). In the full resummed semiclassical amplitude, a one-loop shift of the spectral parameters $\ell_j^{\rm phys} = \ell_j + \hbar\,\delta\ell_j$ shifts the resonance locus by $\delta h = O(\hbar)$. Expanding $(\ell_1^{\rm phys} - \ell_2^{\rm phys})\ln|\ell_1^{\rm phys} - \ell_2^{\rm phys}|/\hbar$ at fixed unshifted $h \to 0$ produces $\delta\ell_{12}\ln h$: the $O(\hbar^0)$ logarithmic divergence (22) is the perturbative signature that the branch point $h = 0$ is shifted by $O(\hbar)$ in physical momentum space **(c)**/**(d)**.

4.2 Physical-region singularities and kinematic stability of the light branch

Theorem 4.3 (Kinematic stability of the light branch and heavy-branch threshold embedding, **(c)**, `phase2/scripts/resonant_channels_quantum.py`).

1. ***Strict concavity and sub-additivity of $E_{\rm light}(P)$:*** *On the light branch $\alpha \in (0,\pi/3)$ ($|P| \in (0,P_{\max})$ with $P_{\max} = \sqrt{3}/8$), $E_{\rm light}(P)$ satisfies $E_{\rm light}(0) = 0$, $\frac{{\rm d}E_{\rm light}}{{\rm d}|P|} = \cos\alpha \in (\frac{1}{2},1)$, and*

$$\frac{{\rm d}^2 E_{\rm light}}{{\rm d}P^2} = -\frac{3}{2\sin 3\alpha} < 0 \qquad \forall\,\alpha \in \left(0,\frac{\pi}{3}\right). \quad (23)$$

By strict concavity on $[0,P_{\max}]$ (for same-sign momenta) and strict monotonicity $|P_1+P_2| < \max(|P_1|,|P_2|)$ (for opposite-sign momenta), any two nonzero light-branch momenta P_1, P_2 with $|P_1+P_2| \le P_{\max}$ obey

$$\boxed{E_{\rm light}(P_1) + E_{\rm light}(P_2) > E_{\rm light}(P_1+P_2).} \quad (24)$$

*Consequently, (E,P) **conservation alone forbids any light soliton from decaying into two light solitons ($1_{\rm light} \not\to 2_{\rm light}$) and forbids any two light solitons from fusing into a light soliton ($2_{\rm light} \not\to 1_{\rm light}$) (c)**. Since every heavy soliton has $M = 2k > \sqrt{3}$, M-conservation ($k_1 + k_2 = k_3$) also forbids $1_{\rm light} \to$ heavy + anything. Thus **light solitons are kinematically stable against all $1 \to 2$ decays (c)**.*

2. ***Every heavy soliton is embedded in the two-light-soliton continuum:*** *Conversely, every heavy soliton S_{13} with $\alpha_{13} \in (\pi/3,\pi/2)$ and direction $\epsilon_{13} \in \{\pm 1\}$ is degenerate in (E,P,M,Π,H_n) with the unique head-on light-light pair S_{12} ($\alpha_{12} = \frac{2\pi}{3} - \alpha_{13} \in (\pi/6,\pi/3)$, $\epsilon_{12} = +\epsilon_{13}$) and S_{23} ($\alpha_{23} = \alpha_{13} - \frac{\pi}{3} \in (0,\pi/6)$, $\epsilon_{23} = -\epsilon_{13}$) on $R_{\rm ho}^{+}$. The 1+1D two-body phase-space integral at the heavy soliton's (E_3,P_3) is non-vanishing:*

$$\int {\rm d}P_1\,{\rm d}P_2\,\delta(P_1+P_2-P_3)\,\delta\big(E_1(P_1)+E_2(P_2)-E_3(P_3)\big) = \frac{1}{|v_{12}-v_{23}|} = \frac{1}{\sqrt{3}\,k_{13}} \in \left[\frac{1}{\sqrt{3}},\frac{2}{3}\right). \quad (25)$$

Across R^{+}_{ho} *(*$\alpha_2 = \frac{\pi}{3} - \alpha_1 - h$*), the on-shell energy denominator* $\Delta E_{\text{on-shell}}(h) := E_1(\alpha_1) + E_2(\alpha_2) - E_{\text{heavy}}(P_1 + P_2)$ *vanishes linearly with non-zero slope:*

$$\frac{\mathrm{d}}{\mathrm{d}h}\Big(E_1(\alpha_1) + E_2(\alpha_2) - E_{\text{heavy}}(P_1 + P_2)\Big)\Big|_{h=0} = -(v_2 - v_3)\,\frac{\mathrm{d}P_2}{\mathrm{d}\alpha_2} \neq 0 \tag{26}$$

(equal to $+0.24107867527655644$ *at* $\alpha_1 = 0.62$*, verified to* 10^{-28} *by finite-step differentiation and* 10^{-58} *analytically in* `phase2/scripts/resonant_channels_quantum.py`*)* **(c)**.

Physical-region singularities of the quantum two-body S-matrix ((c)/(d)). Theorem 4.3 and Proposition 4.1 distinguish two types of physical-region singularities at the resonant loci:

1. **Logarithmic branch point** h^{ich} **in the WKB/Gamma elastic amplitude (c):** In the semiclassical candidate $S_{\text{cand}} \propto \big(\frac{\Gamma(1+\mathrm{i}\eta)}{\Gamma(1-\mathrm{i}\eta)}\big)^{\tau}\big(\frac{\mathrm{e}^2 A_{12}}{\eta^2}\big)^{\mathrm{i}\eta}\mathrm{e}^{\mathrm{i}T_{12}/\hbar}$ with $\eta = \frac{2}{9\hbar}(\ell_1 - \ell_2)$, every resonance line has $\eta = 0$, where $\frac{\Gamma(1+\mathrm{i}\eta)}{\Gamma(1-\mathrm{i}\eta)} = 1$ has **no Gamma pole** (`audit_smx.pdf`). Instead, $A_{12}^{\mathrm{i}\eta} \sim h^{\mathrm{i}\sigma_{\text{res}} c_\ell h/(9\hbar/2)}$ produces a continuous **logarithmic branch point** at real rapidity $h = 0$, across which A_{12} changes sign **(c)**.

2. **Real-rapidity** s**-channel pole / resonance threshold from** $2 \leftrightarrow 1$ **coupling (d):** Because $\Delta E_{\text{on-shell}}(h) = E_1 + E_2 - E_{\text{heavy}}(P_1 + P_2)$ vanishes linearly on R^{+}_{ho} (26), any non-zero quantum $2 \leftrightarrow 1$ transition vertex $g_{12\to3}$ generates an on-shell s-channel propagator pole $\big(E_1 + E_2 - E_3(P_1 + P_2) - \Sigma_3(E, P) + \mathrm{i}\epsilon\big)^{-1}$ directly on the real head-on scattering domain R^{+}_{ho} (and crossed t/u-channel singularities on R^{-}_{ho} and R_{ov}) **(d)**. By Dyson resummation of the two-light-soliton bubble (25), the embedded heavy soliton S_{13} acquires a decay width $\Gamma_{13\to12+23} = \frac{|g_{12\to3}|^2}{2\sqrt{3}\,k_{13}} \geq 0$, turning the real pole into a Breit–Wigner resonance just below the real axis **(d)**.

4.3 Two-soliton unitarity vs. coupled $2 \leftrightarrow 1$ channels

Proposition 4.4 (WKB unitarity violation in the singular domain $A_{12} < 0$, **(c)**, `phase2/scripts/resonant_channels_quantum.py`)**.** *When the semiclassical phase* (19) *(or the continued Gamma candidate) is continued across any of* $R_{\mathrm{ov}}, R^{+}_{\mathrm{ho}}, R^{-}_{\mathrm{ho}}$ *into the singular region* $A_{12} < 0$*,* $\ln A_{12} = \ln|A_{12}| \pm \mathrm{i}\pi$ *acquires an imaginary part* $\pm\mathrm{i}\pi$*. On the physical sheet where the elastic amplitude remains bounded (*$\mathrm{Im}\,\delta > 0$*),*

$$\boxed{\big|S^{\mathrm{WKB}}_{2\to2}(1,2)\big|^2 = \Big|\mathrm{e}^{\mathrm{i}\delta(1,2)/\hbar}\Big|^2 = \exp\!\left(-\frac{4\pi}{9\hbar}\,\big|\ell_1 - \ell_2\big|\right) < 1 \qquad (A_{12} < 0).} \tag{27}$$

For example, at the singular head-on point $(\alpha_1, \alpha_2) = (0.50, 0.65)$ *(*$\epsilon_1 = +1, \epsilon_2 = -1$*, where* $A_{\mathrm{ho}} = -0.354784 < 0$ *and* $|\ell_1 - \ell_2| = 0.295758$*) with* $\hbar = 0.1$*,* $|S^{\mathrm{WKB}}_{2\to2}|^2 = 0.0160837 < 1$*, leaving a two-body unitarity deficit* $1 - |S^{\mathrm{WKB}}_{2\to2}|^2 = 0.9839163 > 0$ **(c)**.

Consequence for unitarity ((c)/(d)).

- **At the resonance curve** R^{+}_{ho} **(**$E_1 + E_2 = E_3(P_1 + P_2)$**):** Because the two-light-soliton state $|P_1, P_2\rangle$ and the heavy one-soliton state $|P_3 = P_1 + P_2\rangle$ share identical conserved charges, unitarity on the energy-momentum shell acts on the coupled two-dimensional space $\mathcal{H}_{E,P} \simeq \mathbb{C}|S_{12}, S_{23}\rangle \oplus \mathbb{C}|S_{13}\rangle$ via a 2×2 unitary block

$$\mathbf{S}(E, P) = \begin{pmatrix} S_{2\to2} & S_{1\to2} \\ S_{2\to1} & S_{1\to1} \end{pmatrix}, \qquad |S_{2\to2}|^2 + |S_{2\to1}|^2 = 1. \tag{28}$$

  Two-soliton unitarity alone ($|S_{2\to2}| = 1$, $S_{2\to1} = 0$) is only consistent if the quantum $2 \leftrightarrow 1$ transition amplitude vanishes identically on R^{+}_{ho} despite the existence of the classical Y-junction trajectory connecting the two states **(d)**.

- **In the singular region $A_{12} < 0$:** Pure two-soliton elastic unitarity fails at the semiclassical level by (27) **(c)**. Because the classical 4-exponential solution blows up ($\tau = 0$) whenever $A_{12} < 0$, two incoming solitons with $A_{12} < 0$ cannot pass through one another as a regular classical two-soliton state; in the quantum theory **(d)**, the absorption deficit $1-\exp(-\frac{4\pi}{9\hbar}|\ell_1-\ell_2|)$ measures the suppression of direct elastic transmission in favour of inelastic/multi-soliton/radiation or resonant rearrangement channels.

4.4 Three-soliton factorisation and bulk fusion bootstrap near resonance

In 1+1D diagonal scattering, the algebraic Yang–Baxter equation $S_{12}S_{13}S_{23} = S_{23}S_{13}S_{12}$ is trivial, but physical three-body factorisation requires that a three-soliton collision decompose into independent pairwise events regardless of chronological order, and—when two solitons fuse ($S_{12} + S_{23} \to S_{13}$)—that scattering a third soliton 3 off the pair before fusion equal scattering off the fused soliton S_{13} after fusion (the bulk fusion bootstrap).

Proposition 4.5 (Classical and bare 1-loop fusion factorisation vs. $O(\hbar^0)$ obstructions, **(c)**, `phase2/scripts/resonant_channels_quantum.py`). *Let $S_{12} = (q_1, q_2)$, $S_{23} = (q_2, q_3)$, and $S_{13} = (q_1, q_3)$ be the three solitons of a KP triple ($q_1 > q_2 > q_3$, sharing $\ell_{\rm res}$), and let $3 = (\alpha_3, \epsilon_3)$ be any test soliton in the regular region ($A(3, S_{ij}) > 0$).*

1. ***Exact classical cross-ratio and Darboux-phase fusion:*** *The shared intermediate root q_2 cancels identically in the product of KP cross-ratios,*

$$\begin{aligned} A(3,S_{12})\,A(3,S_{23}) &= \frac{(p_{3a}-q_1)(p_{3b}-q_2)}{(p_{3a}-q_2)(p_{3b}-q_1)}\cdot\frac{(p_{3a}-q_2)(p_{3b}-q_3)}{(p_{3a}-q_3)(p_{3b}-q_2)} \\ &= \frac{(p_{3a}-q_1)(p_{3b}-q_3)}{(p_{3a}-q_3)(p_{3b}-q_1)} = A(3,S_{13}), \end{aligned} \tag{29}$$

   *and because $M(S_{12})+M(S_{23}) = M(S_{13})$, $\Pi(S_{12})+\Pi(S_{23}) = \Pi(S_{13})$, and $\ell(S_{12}) = \ell(S_{23}) = \ell(S_{13}) = \ell_{\rm res}$, the full Darboux-frame semiclassical phase (19) is **strictly additive across bulk fusion with zero topological remainder**:*

$$\boxed{\delta_{\rm cl}(3,S_{12}) + \delta_{\rm cl}(3,S_{23}) = \delta_{\rm cl}(3,S_{13}).} \tag{30}$$

   *Thus the pure WKB exponential $S = \mathrm{e}^{\mathrm{i}\delta_{\rm cl}/\hbar}$ obeys the bulk fusion bootstrap $S(3,S_{12})\,S(3,S_{23}) = S(3,S_{13})$ identically **(c)**.*

2. ***Pointwise phonon phase telescoping and bare one-loop integral fusion:*** *For every phonon wavenumber $q \in \mathbb{R}\setminus\{0\}$, the one-soliton phonon phase derivatives telescope across the KP triple,*

$$\theta'_{\rm sol}(q;S_{12}) + \theta'_{\rm sol}(q;S_{23}) = \theta'_{\rm sol}(q;S_{13}), \tag{31}$$

   *and therefore the bare fixed-rapidity one-loop continuum phase integrals satisfy $I_{SS}(3,S_{12}) + I_{SS}(3,S_{23}) = I_{SS}(3,S_{13})$ identically **(c)**.*

3. ***Two $O(\hbar^0)$ breakdowns of bulk fusion factorisation:***

   - Maslov/Stirling mismatch in the continued Gamma candidate: *Because S_{12} ($\epsilon_{12} = +1$) and S_{23} ($\epsilon_{23} = -1$) move in opposite directions, their relative orientation signs $\tau(3,S_{12}) + \tau(3,S_{23}) = (+1) + (-1) = 0$ cancel, whereas $\tau(3,S_{13}) = \pm 1 \neq 0$. With common $\eta = \frac{2}{9\hbar}(\ell_3 - \ell_{\rm res})$, the continued Gamma candidate violates bulk fusion at $O(\hbar^0)$ by:*

$$\begin{aligned} \frac{S_{\rm cand}(3,S_{12})\,S_{\rm cand}(3,S_{23})}{S_{\rm cand}(3,S_{13})} &= \left[\frac{\Gamma(1+\mathrm{i}\eta)}{\Gamma(1-\mathrm{i}\eta)}\left(\frac{\mathrm{e}^2}{\eta^2}\right)^{-\mathrm{i}\eta}\right]^{-\tau(3,S_{13})} \\ &= \mathrm{e}^{-\mathrm{i}\frac{\pi}{2}\tau(3,S_{13})\,\mathrm{sgn}\,\eta}\big(1+O(\hbar)\big) \neq 1 \end{aligned} \tag{32}$$

(at $\phi = 0.23, \alpha_3 = 0.31, \hbar = 0.1$, the phase discrepancy is $+1.5158110455271881045$ vs. $+\frac{\pi}{2} = +1.5707963267948966192$ as $\hbar \to 0^+$) **(c)**.

- Fixed-physical-momentum target-soliton renormalisation mismatch: *When $S(3, S_{ij})$ is evaluated at fixed physical momenta $(P_3^{\text{phys}}, P_{S_{ij}}^{\text{phys}})$, the test-soliton Taylor shifts $\Delta X_3(3, S_{ij})\Delta P_3$ cancel by (29), but the target-soliton Taylor shifts fail to telescope because $\Delta P(S_{ij})/k_{S_{ij}}$ varies across the three chords:*

$$\sum_{S\in\{S_{12},S_{23}\}} \frac{\ln A(3,S)}{k_S}\Delta P(S) - \frac{\ln A(3,S_{13})}{k_{S_{13}}}\Delta P(S_{13}) \neq 0 \tag{33}$$

(equal to $+0.8755292547543045976$ at $\phi = 0.23, \alpha_3 = 0.31$) **(c)**.

Remark 4.6 (Wave-packet / Landau breakdown of 3-body chronological independence near resonance, **(d)**). Classically, (29)–(30) hold because the fused intermediate soliton S_{13} carries the exact sum of the KP chords of S_{12} and S_{23}. Quantum mechanically **(d)**, near the resonance locus ($|h| \ll 1$), the divergent time delay $\Delta t_{12} \sim \frac{1}{|v_1 - v_2|k_1} \ln(1/|h|)$ means that in a 3-body wave-packet collision with impact parameter separation $\Delta X \sim v\,\Delta t_{12}$, the intermediate state is an on-shell coherent superposition of $|S_{12}, S_{23}\rangle$ and $|S_{13}\rangle$. Because the $O(\hbar^0)$ scattering phases off $|S_{12}, S_{23}\rangle$ vs. $|S_{13}\rangle$ differ by the nonzero target-momentum renormalisation (33) (unless cancelled by a two-body quantum frame counterterm), chronological independence of $3 \to 3$ scattering fails at $O(\hbar^0)$ in the resonant transition zone **(d)**.

5 Question 4: Conclusion—when is the quantum S-matrix elastic and factorised?

Combining our exact structural results **(c)** with their quantum implications **(d)**, we answer Question 4 by partitioning the two-soliton kinematic space into four physically distinct regimes:

1. **Regime I: Light-light overtaking collisions** ($\alpha_1, \alpha_2 \in (0, \pi/3)$, $\epsilon_1 = \epsilon_2$).
   Status: **Elastic and factorised** (**(c)** for kinematic protection, **(d)** for full quantum theory).

   - Every soliton has $|P_j| < P_{\max} = \sqrt{3}/8$ on the KdV-like light branch, $\alpha_1 + \alpha_2 < 2\pi/3$ everywhere, so $0 < A_{\text{ov}} < 1$ is strictly regular and bounded away from R_{ov} except at the double-cusp corner $(\pi/3, \pi/3)$ **(c)**.
   - By Theorem 4.3(1), $1_{\text{light}} \to 2$ decay and $2_{\text{light}} \to 1_{\text{light}}$ fusion are strictly forbidden by (E, P) conservation, and $2_{\text{light}} \to 1_{\text{heavy}}$ overtaking fusion is forbidden by (M, P, E) conservation ($\alpha_1 + \alpha_2 < 2\pi/3$) **(c)**.
   - No resonant or number-changing classical channel is reachable; a unitary, diagonal, factorised quantum two-soliton S-matrix is fully consistent here **(d)**.

2. **Regime II: Sub-threshold light-light head-on collisions** ($\alpha_1 + \alpha_2 < \pi/3$, $\epsilon_1 = -\epsilon_2$).
   Status: **Elastic below the heavy-soliton threshold** R_{ho}^+ (**(c)**/**(d)**).

   - In this triangle ($\frac{1}{2}$ of the light-light square, $\frac{2}{9}$ of Q), $A_{\text{ho}} > 1$ is strictly regular and attractive **(c)**.
   - The total energy lies strictly below the heavy-soliton threshold: $E_1(\alpha_1) + E_2(\alpha_2) < E_{\text{heavy}}(P_1 + P_2)$ (since $\Delta E_{\text{on-shell}}(h) > 0$ for $h = \pi/3 - (\alpha_1 + \alpha_2) < 0$ and < 0 for $h > 0$ when $\alpha_1 > \pi/6$, or vanishes only on R_{ho}^+), and (M, P, E) conservation forbids $2 \to 1$ fusion anywhere off R_{ho}^+ **(c)**.
   - As $\alpha_1 + \alpha_2 \to (\pi/3)^-$, the collision approaches the on-shell heavy-soliton resonance R_{ho}^+, where the classical delay diverges as $\ln(1/h)$ and the s-channel heavy-soliton pole/branch point sits on the boundary of the elastic domain **(c)**/**(d)**.

3. **Regime III: Heavy-soliton channels ($\alpha_j \in (\pi/3, \pi/2)$) and the resonant loci $R^+_{\rm ho}, R^-_{\rm ho}, R_{\rm ov}$.**
Status: **Inelastic $1 \leftrightarrow 2$ and resonant rearrangement channels open** (**(c)** for kinematics/conservation, **(d)** for quantum amplitudes).

   - Every heavy soliton S_{13} ($\alpha_{13} \in (\pi/3, \pi/2)$) is an embedded state in the two-light-soliton head-on continuum with exact conservation of M, Π, P, E and all H_n into $S_{12}+S_{23}$ and non-zero 1+1D phase space $J_{1\to 2} = (\sqrt{3}\, k_{13})^{-1}$ **(c)**. Unless a hidden quantum selection rule forces the $1 \leftrightarrow 2$ matrix element to vanish identically, heavy solitons are unstable quantum resonances ($S_{13} \to S_{12} + S_{23}$) rather than stable asymptotic particles **(d)**.
   - Semiclassically, near $R^+_{\rm ho}$ ($h = \pi/3 - (\alpha_1 + \alpha_2) \to 0$), the $2 \to 2$ amplitude couples to the $2 \to 1$ fusion channel via the on-shell energy denominator $\Delta E_{\text{on-shell}}(h) \propto h$, while across the resonance curve ($h < 0$, entering $A_{12} < 0$) the direct WKB elastic transmission amplitude is suppressed by the barrier-penetration factor **(c)**/**(d)**:

$$\left|S^{\rm WKB}_{2\to 2}\right| = \exp\!\left(-\frac{2\pi}{9\hbar}|\ell_1 - \ell_2|\right), \qquad P^{\rm WKB}_{\rm inelastic} = 1 - \left|S^{\rm WKB}_{2\to 2}\right|^2 = 1 - \exp\!\left(-\frac{4\pi}{9\hbar}|\ell_1 - \ell_2|\right). \tag{34}$$

4. **Regime IV: Open singular two-soliton domains ($A_{12} < 0$) and complex Hirota breathers ($A_{\rm br} < 1/4$).**
Status: **Non-elastic / outside the regular soliton sector** (**(c)**/**(d)**).

   - For $\alpha_1 + \alpha_2 > 2\pi/3$ (overtaking) and $\alpha_1 + \alpha_2 > \pi/3$, $|\alpha_1 - \alpha_2| < \pi/3$ (head-on), $A_{12} < 0$: the classical 4-exponential solution develops $\tau = 0$ poles, and the continued WKB elastic amplitude violates $|S_{2\to 2}| = 1$ by (27) **(c)**. Even in the light-light head-on square $(0, \pi/3)^2$, the upper triangle $\alpha_1 + \alpha_2 \in (\pi/3, 2\pi/3)$ lies in $A_{\rm ho} < 0$, so **even restricting external states to the light branch $\alpha < \pi/3$ does not make head-on scattering globally elastic for $\alpha_1 + \alpha_2 > \pi/3$ (c)/(d)!**
   - Every two-phase zero-background Hirota breather of the good equation has $A_{\rm br} < 1/4 < 1$ and is classically singular (`phase1/review.pdf`, `audit_smx.pdf`), so there is no regular classical breather family to require Bohr–Sommerfeld bound-state poles on the physical sheet **(c)**/**(d)**.

6 Summary of claims, labels, and verification scripts

All three verification scripts in `phase2/scripts/` run in under 25 seconds total and terminate with `ALL CHECKS PASSED`.

Claim / Formula	Label	Script	Precision / Check
KP chord dictionary (3)–(5), cross-ratio = Hirota = trig forms	**(a)**/**(c)**	`phase2/scripts/resonant_channels_kinematics.py`	Exact symbolic over $\mathbb{Q}(\sqrt{3})[t_1, t_2]$
Exact resonance lines $R_{\rm ov}, R^+_{\rm ho}, R^-_{\rm ho}$; refutation of $\|\alpha_1 - \alpha_2\| = \pi/3$ for overtaking	**(c)**	`phase2/scripts/resonant_channels_kinematics.py`	60-digit mpmath ($< 10^{-48}$)
KP-triangle unification (10)–(11): 2 light chords + 1 heavy chord	**(c)**	`phase2/scripts/resonant_channels_kinematics.py`	Symbolic + 60-digit mpmath
Exact 3-exponential Y-junction PDE solution and $1 \leftrightarrow 2$ asymptotic limits	**(a)**/**(c)**	`phase2/scripts/resonant_channels_kinematics.py`	60-digit PDE jet ($< 10^{-50}$) and $t = \pm 400$ ($< 10^{-48}$)

Claim / Formula	Label	Script	Precision / Check
Kinematic area fractions ($\frac{7}{9}$ reg. ov., $\frac{1}{3}$ reg. ho., 100% light-light ov.)	**(c)**	`phase2/scripts/resonant_channels_kinematics.py`	Exact rational geometry
Symbolic integration of $H_1, \ldots, H_8$ via $T = \tanh(kx/2)$ (14)	**(c)**	`phase2/scripts/resonant_channels_conservation.py`	Exact symbolic polynomial integration
Univariate chord potentials $F_Q(p)$ (15) for M, Π, P, E and $H_1, \ldots, H_8$	**(c)**	`phase2/scripts/resonant_channels_conservation.py`	Exact polynomial division mod $p_a^2 + p_a p_b + p_b^2 - \frac{1}{4}$
Spin-3 gap: pure $k^3 \notin \mathrm{span}\{\Delta_1, \Delta_5\}$, while $E \in \mathrm{span}\{\Delta_1, \Delta_5\}$	**(c)**	`phase2/scripts/resonant_channels_conservation.py`	Exact symbolic linear system
Classical telescoping $Q(S_{12}) + Q(S_{23}) = Q(S_{13})$ and Y-junction quadrature across $t \in \{-2, 0, 2\}$	**(c)**	`phase2/scripts/resonant_channels_conservation.py`	Exact symbolic + 60-digit quadrature ($< 10^{-50}$)
One-loop mode-sum telescoping vs. fixed-P mismatch $\Delta(F\|_P) = 0.3887924407\ldots$ (18)	**(c)**	`phase2/scripts/resonant_channels_conservation.py`	60-digit mpmath ($< 10^{-48}$)
Transverse $h \ln h$ branch point (21) and fixed-P $\ln A_{12}$ divergence (22)	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	60-digit asymptotic extraction ($< 10^{-18}$)
Strict concavity $\frac{\mathrm{d}^2 E_{\mathrm{light}}}{\mathrm{d}P^2} = -\frac{3}{2\sin 3\alpha} < 0$ and $1_{\mathrm{light}} \not\leftrightarrow 2_{\mathrm{light}}$ (24)	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	Exact symbolic + 60-digit grid check
Heavy-branch $1 \to 2$ Jacobian (25) and linear on-shell energy zero (26)	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	60-digit analytic + finite-step ($< 10^{-28}$)
WKB unitarity deficit $\lvert S_{2\to 2}^{\mathrm{WKB}}\rvert^2 = \mathrm{e}^{-\frac{4\pi}{9\hbar}\lvert \ell_1 - \ell_2\rvert} < 1$ for $A_{12} < 0$ (27)	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	60-digit evaluation
Classical bulk fusion $A_{3,12} A_{3,23} = A_{3,13}$ and $\delta_{\mathrm{cl}}(3, S_{12}) + \delta_{\mathrm{cl}}(3, S_{23}) = \delta_{\mathrm{cl}}(3, S_{13})$	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	60-digit mpmath ($< 10^{-48}$)
Pointwise phonon telescoping (31); Gamma candidate $\mathrm{e}^{\mp \mathrm{i}\pi/2}$ and target-ΔP obstructions	**(c)**	`phase2/scripts/resonant_channels_quantum.py`	60-digit mpmath ($< 10^{-48}$)
Physical elasticity conditions, heavy-soliton decay width, and inelastic regime partition	**(d)**	—	Conditional on quantum vertex and frame map

7 Failures recorded during development

In accordance with `AGENTS.md`, we record every failed check or error encountered during the development of this package and how it was resolved without weakening any check:

1. **SymPy trigonometric cube simplification and rational GCD hang in `phase2/scripts/resonant_channels_kinematics.py`:** In the initial version of `phase2/scripts/resonant_channels_kinematics.py`, `sp.simplify(sp.expand_trig(4*p_root**3 - p_root - ell1/(3*sp.sqrt(3))))` returned a non-zero unexpanded expression containing `sin(alpha_1 + pi/6)**3`, and `sp.cancel` over $\mathbb{Q}(\sqrt{3})[t_1, t_2]$ hung on rational cross-ratio differences. Clearing polynomial denominators using the homogeneous Weier-

strass parametrization $S_j = 2t_j$, $C_j = 1 - t_j^2$, $D_j = 1 + t_j^2$ reduced all identities to pure polynomial expansions in $\mathbb{Q}(\sqrt{3})[t_1, t_2]$ that vanish identically in $< 0.1\,\mathrm{s}$.

2. **Trailing-digit fabrication on the false overtaking resonance value in `phase2/scripts/resonant_channels_kinematics.py`, and insufficient asymptotic time $T = 60$:** In `phase2/scripts/resonant_channels_kinematics.py`, an initial assertion compared $A_{\mathrm{ov}}(0.2, 0.2 + \pi/3)$ to a 20-digit string obtained by appending arbitrary digits to the 12-digit number 0.355465479849 quoted in `audit_smx.tex`; the assertion failed because the true 25-digit value is 0.35546547984906164961877 23. It was replaced by an exact 10^{-48} check against the closed form $\frac{1-3\cot^2(\alpha_1+\pi/6)}{-8}$ together with a 10^{-12} check against the rounded audit figure. In the same script, checking the $t \to \pm\infty$ peak height of the Y-junction at $T = 60$ failed the 10^{-25} tolerance because the transverse exponent $\sqrt{3}k_{12}k_{23}T \approx 17.3$ gives $\mathrm{e}^{-17.3} \approx 3\times10^{-8}$; evaluating at $T = 400$ ($\mathrm{e}^{-115} < 10^{-49}$) passed the stricter 10^{-48} tolerance.

3. **Initial mental guess for the rational coefficients of H_7 and H_8 in `phase2/scripts/resonant_channels_conservation.py`:** Before running the symbolic integrator in `phase2/scripts/resonant_channels_conservation.py`, the docstring contained a tentative hand-estimated decomposition for H_7 and H_8. Exact symbolic integration of $\mathrm{Res}\, L^{7/3}$ and $\mathrm{Res}\, L^{8/3}$ via $T = \tanh(kx/2)$ yielded the unique exact coefficients $H_7 = \Delta_7 - \frac{7}{12}\Delta_5 + \frac{7}{324}\Delta_1$ and $H_8 = \Delta_8 - \frac{7}{36}\Delta_4 + \frac{11}{324}\Delta_2$, which pass exact polynomial division modulo $p_a^2 + p_a p_b + p_b^2 - \frac{1}{4}$.

4. **Unbracketed Newton overshoot across the cusp $\alpha = \pi/3$, sign of $P_1 + P_2$ on R_{ho}^+, and an extra $\sin\alpha$ factor in $\mathrm{d}P/\mathrm{d}\alpha$ in `phase2/scripts/resonant_channels_quantum.py`:** In the first run of `phase2/scripts/resonant_channels_quantum.py`, inverting $P(\alpha) = P_{\mathrm{target}}$ on the light branch by unbracketed Newton iteration from $\alpha_0 = 0.5$ overshot the cusp $\alpha = \pi/3$ (where $P'(\pi/3) = 0$) onto the heavy branch when $P_1 + P_2$ was near $P_{\max}$, causing the sub-additivity assertion $E_{\mathrm{light}}(P_1) + E_{\mathrm{light}}(P_2) > E_{\mathrm{light}}(P_1 + P_2)$ to fail. Replacing unbracketed Newton by bracketed bisection on $[0, \pi/3]$ (light) and $[\pi/3, \pi/2]$ (heavy) followed by Newton polishing resolved the branch jump. In the same test, choosing $\alpha_1 = 0.42 < \pi/6$ ($\epsilon_1 = +1$) and $\alpha_2 = \pi/3 - 0.42 > \pi/6$ ($\epsilon_2 = -1$) gave $P_1 + P_2 < 0$, so the fused heavy soliton had $\epsilon_3 = -1$ rather than $+1$; and the Newton polishing step initially wrote $\frac{\mathrm{d}P}{\mathrm{d}\alpha} = \frac{2}{3}\sin^2\alpha\sin 3\alpha$ instead of the correct $\frac{2}{3}\sin\alpha\sin 3\alpha$, which degraded the finite-step derivative check of $\Delta E'_{\text{on-shell}}(0)$. Correcting $\frac{\mathrm{d}P}{\mathrm{d}\alpha}$ and testing $\alpha_1 = 0.62 \in (\pi/6, \pi/3)$ ($P_1 + P_2 > 0$, $\epsilon_3 = +1$) passed at 10^{-28} (finite-step) and 10^{-48} (analytic).

Reference verification and reading scope. All references below are inherited from the verified repository bibliography (`phase2/report.tex` and `phase2/errata.tex`); no new external reference is introduced. Reading scope follows the project ledger: relevant full-text sections read in the project for [1, 2, 3, 5, 6, 9]; abstract/metadata verified in `errata.tex` for [4, 7, 8, 10].

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